Supplementary Table 1 Characteristic of worksite physical activity (PA) interventions delivered to older employees (organised by publication dates)

Study	Recruitment / reach	Intervention content / theoretical guidance	Intervention deliverers	Duration	Process data
Bassey, E. J., et al, 1987 [27]	Initial discussions for the support of the management; After a publicity launch in the factory newspaper, a personal letter of invitation was sent to all workers through factory's internal mail system; 35% attended introductory meetings and ultimately 23% agreed to participate	"The New Aerobics" (1970) Cooper's prescription for people aged ≥50yrs, which isa progressive walking program instructed at workplace but carried out at employees' leisure time. 1st week, walking 20mins$_{*}^{-1}$d and at a pace of 5.2km$_{*}^{-1}$hr preferably 5 days per week. Subjects can increase duration of walks and frequency. In 2nd phase the intensity can increase based on heart rate response to exercise at 4.8km$_{*}^{-1}$hr. The target dose was 40mins at 7km$_{*}^{-1}$hr 5 days per week. Individual discussion at beginning (how to progress) and half way through. The progression was individually discussion on courses and timing at start and mid program. Participants were asked to keep daily log-card of duration and distance walked. Encouraging articles in the factory newspaper were published monthly.	Unsupervised program – no mentioning of the profession of the person conducted the meeting to inform the grade of effort	12 weeks program divided to 6 weeks each phase	59 out of 108 completed the intervention (54%). Log cards: only 1/3 nearly achieved the speed and distance prescribed for the last 2 weeks, 1/3 provided evidence on logs for extra walking but substantially below prescription, and 1/3 failed to return log card. The most compliant were the most active before program started
Sharpe, P. A. et al 1992 [35]	Advertising at work, direct contact with departmental representatives, telephone calls from the research staff and provision of additional dates and sites to enhance the accessibility of screening; 250 employees did health screening but the % from total employees is unknown	Health promotion activities after health risk screening (smoking, blood pressure, level of blood lipids and body weight) and brief communication of results by nurse. Participants could choose to have face-to-face with meetings with health counsellors, to join walking groups at work and/or exercise on their own at home or at university facilities. No other promotional activities were reported.	health counsellors at work provided face-to-face with counselling based on participants choice	12 months	60 out of 95 experimental participants (63%) completed the program. Behaviour change self-report: 48% of Control and 53% of have done anything to increase fitness. started program on their own- 63% and 73% of C and Int. took worksite program 13% of C and 11% of Int; used worksite facilities: 11% of C 16% Int.
Arao, T. et al 2007 [36]	Two companies were recruited, all together 5 sites. Medical check-ups were conducted and 192 employees were identified to have CVD risk factor(s) of those 177 (92%) were eligible and consented and ultimately 155 (87%) were allocated as a cluster to LiSM (n=92) and Control (n=85) groups.	Theoretically driven (Transtheoretical model and Social Cognitive Theory) Life Style Modification Program for Physical Activity and Nutrition (LiSM-PAN). Structured counselling for goals setting for each behaviour and monthly meeting. Log book for steps a day (10,000 as a goal); increasing incrementally; if reached take another sport/exercise class. Environment support at worksite: 1) setting up a walking course; 2) installing exercise facilities; 3) displaying caloric content table of the lunch menu at cafeteria; 4) cafeteria manager advised on healthy cooking for better diet. Social support at home: printed material on healthy diet and cooking; participant and spouse going together for exercise	Research staff were trained in structured counselling;	6 months	95.2% completion. 97.1% attended monthly counselling 86.7% self-monitored the walking steps 54.7% controlling their dietary targeted activities

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Granacher et al. 2011 [34]	Several companies in Basel were approached and 2 large companied (~ 500 employees) agreed (randomization at cluster level). Informational session was held explaining the intervention and eligibility criteria, 34 employees consented and participated.	Three times a day five days a week of training in: Static balance exercises – in morning at arrival to office Dynamic balance exercises – after lunch break Strength enhancing exercises before leaving office New exercise routines were assigned each week to be repeat for the whole week before progressing.	Expert in balance and strength training saw each participant every week	8 weeks of training followed by 8 weeks without training data collected at 8 and 16 weeks	All 17 intervention participants completed the program (100%). Program compliance was excellent with participants of the intervention group completing 99% of the scheduled training sessions (~ 119 out of 120 training sessions).
Hughes et al. 2011 [24]	Worksite advertisement Research participants recruited via announcements on staff list-serves, targeted e-mails, staffed recruitment tables at events in highly trafficked buildings, and flyers posted throughout the university. Out of 647 enquiries 66% deemed eligible and 423(65%) enrolled in the study.	COACH –shaped by the Enhance Wellness Program for seniors in the community and guided by the Transtheoretical Model. One face- to- face meeting with coach to review the participant's health-risk profile (baseline assessment), discuss the behaviors the participant was interested in adopting, and discuss the availability of health behavior change programs on or off campus, assist in creating plan to achieve goals. Biweekly contact by email/telephone for the first 6 months to re-evaluate plan, goals and actions. In the remaining 7 to 12 months the contact was monthly. Health-risk profile assessments were at 6 and12 months. RealAge- A Web page contain health risk test and generate health profile based on family history, diet and nutrition, PA, tobacco and alcohol use, satisfaction with family relationships, and use of preventive health care. Participants used the Web site to select behaviors to work on and create plans to meet behavioral goals. The RealAge site tracked each time a participant used it, and the site forwarded this information to the study team at regular intervals, enabling us to document use.	COACH -MPH graduate trained in behavioural change coaching RealAge – research staff	12 months	Retention: 91% in COACH and 84% RealAge. Fidelity data: COAC: Coach spent 9.25hr per participant – there were 688 emails 77% by the coach and 48 telephones FU, RealAge- 59% completed all health risk assessment and 53% engaged in at least one of the web link.
Palumbo et al. 2012 [26]	One medical centre targeting nurses currently employed full time or part time in staff nurse position that involved lifting patients. About 70 nurses place their interests and 14, on a first come first served basis, were randomized.	Tai Chi workplace wellness program, guided by the Revised Health Promotion Model by Pender (1996). Classes on site were carried once a week (including 15 minutes breathing exercise, 30 mins tai chi practice 5mins visualization cool down). Additional practice on their own 10mins 4 times a week	Experienced tai chi instructors (22 years of teaching)	15 weeks	Five out of the 7 intervention participants (71%) completed the tai-chi program. Attendance 82% range:69%-92% Fidelity- two classes were cancelled

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Strijk et al. 2011 , 2012 [22,23]	All workers of two academic hospitals were invited to participate between April and October 2009; Among 3756 employees invited to participate, 1101 (29%) agreed to participate and 730 (19%) allocated into two groups	3 visits to Personal Vitality Coach (PVC) at worksite: at start of intervention, followed by 2 consecutive visits at 4-6 weeks and 10-12 weeks. Meeting 30 minutes each. Weekly yoga 45mins and workout sessions (intensity 65%-90% of the age-predicted maximum heart rate) for 45mins. Plus 45 min unsupervised aerobic exercise. Fruit & Vegetable at the session program name 'Vital@work'	Certified yoga instructor and exercise physiologist	6 month interventio n period and additional 6 months sustainabili ty testing	293 out of 367 (80%) intervention participants completed the program. PVC visits 100%. Dose delivered – Yoga sessions 72.3% (Amsterdam - 89.3%; Leiden - 58.3%) Workout sessions 96.3% (Amsterdam – 95.1%; Leiden – 97.4%) High satisfaction score
Cook, et al. 2015 [27]	A recruitment flyer was emailed by company officials to all employees 50 years plus; interested employees were provided information on study and a link to the online baseline survey (received $25 if completed); out of 290 employees, 279 (96%) completed the baseline survey.	HealthyPast50, an automated web-based program containing multimedia health promotion information and guidance based on Social Cognitive Theory. Modules: Facts about Healthy Ageing, My Health Profile – assessment module with questionnaires and recommendations, Stress and Mood Management, Healthy Eating, Active Lifestyle, and Tobacco Free. Encouragement of self-assessment (My Health Profile) before accessing modules. the content of the Active Lifestyle module is oriented toward older users, including the use of the Physical Activity Readiness Questionnaire (PAR-Q), assessing one's readiness for physical activity, the relationship of physical activity to mobility and energy in later years, and the need to moderate the intensity of exercise as one ages	No human contact for the program; project staff can be reached by telephone for queries	3 months; program progress by monthly email and reminder to access	Formative evaluation of the program including focus groups and testing materials. Process evaluation: mean # log-in:4.33 (SD 4.28, range 0-28), the mean minutes in the program was 102.26 minutes (SD 148.32), and the mean number of pages viewed was 11.04 (SD 20.08, range 0-120).
Abbas et al. 2015 [38]	Targeted recruitment of low-paid unskilled or semi-skilled employees age >40 on social care. A single individualised invitation was sent with simultaneous promotion of the programme within standard employer–staff communication routes. Out of 3264 target group (low SES employees), 665 (20% response rate) attended health screening. Of them, 579 were eligible and 427 (74%) returned for the follow-up assessment	Health screening of all major cardiovascular risks was offered to employees of low SES in deprived area. Based on screening results participants were stratified to either "high" , "moderate" or "low" CVD risk. The intervention included referrals to either GPs (if pharmacological treatment were needed), Weight management (in the community, exercise training, smoking cessation and alcohol reduction counselling either face to face or by telephone. Employers ensured that time of assessments and time spent on health promotion will be accounted towards the daily working hours. These services were provided in the community.	Occupational health nursing staff. The nurses were trained to provide smoking cessation and alcohol awareness delivery at the national standard.	9 months interval from initial assessment to follow-up	The attendance to health screen was 20%, much lower than the general population that offered similar program (health screen). Fidelity=There was a record of the referral number given to each domain of intervention (exercise 38%, smoking 5%, alcohol 1%and weight management 20%). No account on the attendance to service referred.

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Low et al. 2015 [28]	A promotional flyer to all female at age range & invited to call the centre if interested in reduction of CVD risk; once reached 60 employees who had ≥1 risk of CVD no further recruitment effort was made	Multi-components risk reduction educational sessions in groups on: weight loss/nutrition, stress management, exercise training, and smoking cessation accompanied by access to an on-site gymnasium, and organized group walks. Intervention groups received weekly communication via phone or e-mail, incorporating goal-setting and suggestions for overcoming obstacles.	Not specified	6-months for classes, gym availability organized walks and weekly communication Follow-up survey at 12 months	Attendance to 1st classes was highest for weight loss (42% & 47% control). Stress reduction (30% & 21%) and physical activity 25% to 26%) followed by very poor attendance to consecutive classes. 3 attended group walking; use of Gym 0-20 days over 6 months . Follow-up 16/57 (28%)
Chen et al. 2016 [37]	Approached six worksites small- and medium scale enterprises (SMEs). Three sites showed interest and approved. Among 1245 workers in the three sites, 438 (35.2%) met study criteria of being full-time employed and aged over 50 years. These were invited. Of those, 264 (60.3%) showed interest and finally 108 (25%) agreed to participate.	Phase I: on-site lectures and workshops on healthy living specifically impact of metabolic disorders and individualized self-management goal setting. Biweekly meetings face-to-face with occupational health nurse (OHN) to share experience, give mutual support and encouragement, learn about managing own health, and cultivate leadership. Phase II: on-site team led meetings; discussing barriers reinforcing personal goals; recording food consumption and exercise. Logs were reviewed by OHN and awards/prizes were given for excellent improvement	Occupational Health Nurse	Total 24 weeks Phase 1 was 4-week phase II 20- weeks.	111 from initial 136 intervention participants (80%) completed the program Qualitative Evaluation: Participants found it difficult to records daily PA and diet so no review of records was made; Working overtime reduce the compliance with planned activities and goals; activities forced to cut short due lack of support from management at low level
Naug et al. 2016 [39]	Contacted with employer; Information session meetings were arranged to build rapport with the employers and drivers; after that, drivers were invited to an information presentation and were given an information package.	Educational sessions of 60mins delivered fortnightly in groups at worksite, interactive and fun. Content were derived from the "Life! Taking action on diabetes" program which was based on health psychology theories with a focus on self-regulation. Targeting physiological change due to inactivity, harm of sitting, healthy nutrition. Participants were provided pedometers and asked to record steps each day for 6 weeks.	Two postgraduate students with expertise in adults' education, human physiology & nutrition	6 weeks	21 out of 33 that started program (64%) completed the program. Qualitative after intervention : drivers said " I walk around the bus during breaks" "I, use the back seats of the bus to do push-ups"; "park my car far from depot to walk more"

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Kocur et al. 2017 [29]	Workplace advertisement Postmenopausal female who replied to the advertisement regarding the study (n=91) and net the inclusion criteria (n=44) but no data on % from the total number of the target population.	Nordic walking training program techniques and training 3 times a week for one hour including 10-min warm up phase at start 5-min cool down at end. Intensity progress 40%-70% of heart-rate reserve	Two instructors specialised in Nordic walking regulated the pace	12 weeks	No report on program attendance or fidelity.
Chopp-Hurley et al. 2017 [31]	Workplace advertisement through e-mail list serves, webpage at McMaster workplace wellness committee Potential participants had to have clinical osteoarthritis either of the hip or the knee; no number was provided for those that were screened or refused.	Program of static lower extremity strengthening exercise; designed initially for Knee Osteoathritis to avoid KAM (knee adduction moment); The program includes Yoga poses such as squats and lunges; A perceived intensity of 5-7 from a total 10-point scale was recommended. On-site 4 timea a week sessions in the University sport and recreation facility were provided before work (7-8 AM). Participants were asked to attend at least 3 sessions/week. Participants in intervention received weekly feedback on their performance of exercises. Incentive of $50 was given to maintain participation in both groups and control participants were offered the program at the end of trial	Not specified	12 weeks	From initial 24 participants (12;12) 3 intervention participants did not complete FU (25%) as opposed to 1 in the control (12%). Exercise group attended on average 1.2 sessions per week; median was 0.5 classes per week, lower quartile was 0 a week and upper quartile was 3.9 per week thus estimate that 33% adhered to 3 sessions per week.
Bergman et al. 2018 [32]	13 different companies in 17 sites were involved (number of employees age 40-67 not given). Information session were given to office workers-the information included who is eligible (i.e., at least 4 days/wk in the office with sedentary work tasks with a sit-to-stand workstation, don't have a travel day during their work). Information on study strict procedures and measurement only 100 assessed and 80 enrolled	Health consultation about 30-40 minutes about their screening results and sitting time at work and diet. Folder with written information about PA recommendations for moderate and vigorouse aerobic activity and strength, how to increase light PA and Non-exercise PA such as climbing stairs, walk and talk meetings. After this session the treadmill adapted to desk use was installed next to the existing workstation. Physiotherapist adjust the workstation and instruct the worker how to use the treadmill in the first week – any pace 15min twice a week. Thereafter they were recommended to use it for 1hr a day preferably longer. Booster email from research team remined the hazard of sitting, reminder to use treadmill at least 1hour a day. These emails were sent at 6,20,31 and 50th week over the program.	Nurses conducted the health consultation and boosters emails by research team	13 months	Of the 80 randomised (40;40) 34 continued their participation for 13 months. In the control 92% continued for 13 months. No data was collected on usage of the treadmill during work Threat for contamination between group was acknowledged

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Scapellato et al. 2018 [40]	Targeted strategy; screening for eligibility (at least 1 CVD risk: blood pressure, weight, waist circumference lipids, impaired fasting glucose) 1892 workers on voluntary basis agreed to be screened; 1574 did not meet the inclusion criteria 318 recruited (50% from 632 eligible). Final recruitment 226 (35% from eligible).	As part of the Occupational Health and Safety approach, a workplace intervention was offered to promote healthy lifestyle for healthcare workers in the hospital who are at risk; intervention included individualized counselling meeting a month after baseline assessment to discuss results of PA and diet, and diverted to health counselor to identify PA and diet goals and strategies to achieve them. Helpful printed material were given such as food pyramids and a brochure of exercise. After 3 months participants received further motivational support via phone counselling to remind their goals and assist to change lifestyle habits.	Hospital Employees Occupational Health practitioners, health promotion counsellors and hospital dietitians	6 months	Fidelity=92 out of 318 eligible recruited (29%) did not receive the intervention. Of the 226 that received intervention, only 169 remained for 6-month assessment. No information on % phone calls given at 3-month. Reason to lost to follow up were attributed to “no time” & “too many work commitments”; only 40% who completed the study favoured adherence to dietary recommendations
Qi et al. 2019 [33]	Worksite advertisement; posters, person-to-person invitations using card-sized flyers, promotion of the study by colleagues and an e-mail request from the University Administration promoting the research projects that office workers may be interested to partake in. No data on number of potential eligible (55+ in the Uni) or how many were interested	Based on evidence that taichi or resistance training alone can improve physical function and psychological wellbeing of older adults. The intervention offered 3 session per week from a choice of 8 different timetables to allow flexibility: Monday, Wednsday and Fridays (2 session) 7-9 AM midday and afternoon 3-6PM. Session took 45 minutes. Ten Taichi movements were practiced with theraband tied to palms to a relaxing/ calming music. Theraband degree of effort were progressive. Every fortnight a Rating of Perceived Exertion was completed. When RPE was below 5 the degree the difficulty was increased	Taichi instructor, which is the first author of the paper, delivered the intervention	12 weeks with mid-term (6-weeks) assessment	At 6 week 9 participants withdrew (22%) and no further withdrawals happened at 12 weeks. Reason to lost to follow up were: back issue, knee issue, time conflict flue and no reason given. Data on compliance was not reported; there were 36 sessions offered but no data on mean session attend.
Edman et al 2019 [41]	Worksite advertisement; e-mail invitations (3) to get health coaching for free. Flyers were posted in various locations in the main hospital including the cafeteria and in satellite office locations.	Health coaching on cardiovascular health delivered face-to-face or by phone or in combination. A cyclie is considered 12 weeks during which 6 consultations 2-30min each must be completed. Sessions are client centred behavioural changes strategies and include provision of education and individualized support to make change. Sessions start with assessment, changes from previous session, goal setting (fortnightly), elaborate on topic the client chose (the SMART framework model)	A certified Health Coach with 4 years’ experience with a PhD degree in Nutrition (the first author)	5.5 months with cycle is 12 weeks with option to continue.	74% (40/50 participants) completed 12 weeks. Participants averaged 7.8 sessions (range 6–12) over 5.5 months (range 3.0–10.0). Coaching were in-person for 57% and 17% receive only telephonic coaching and 26% received combination of both.